



AIR POLLUTION FROM ENERGY USE IN A DEVELOPING COUNTRY CITY: THE CASE OF KATHMANDU VALLEY, NEPAL†

RAM M. SHRESTHA‡ and SUNIL MALLA

Energy Program, School of Environment, Resources and Development, Asian Institute of Technology,
 G.P.O. Box 2754, Bangkok 10501, Thailand

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Abstract—We have examined sectoral energy-use patterns and estimated the associated emission of key air pollutants in Kathmandu Valley for 1993, as well as the levels of pollutant emissions from fuel use in 2013 under a business-as-usual (BAU) scenario; total emissions of selected pollutants are estimated to be over 63,000 tons in 1993 and to increase five-fold by 2013. The transport sector contributes the largest share of pollutants, followed by the household and industrial sectors. Gasoline, fuelwood and coal were the dominant fuel contributors to total emissions and CO was the dominant pollutant. Copyright © 1996 Elsevier Science Ltd.

1. INTRODUCTION

Kathmandu Valley (approximately 300 km²) is the largest urbanized area and the capital city of Nepal with a population of 1.1 million (1991). The city is known as the City of Temples and is the treasure house of Nepalese culture, arts, and architecture. In recent years, air pollution has been emerging as a major environmental problem. It is now not uncommon to see people wearing face masks or putting shawls or handkerchiefs over their mouths and noses. The valley is especially vulnerable to air pollution due to its bowl-like topography, which restricts wind movements and retains air pollutants in the atmosphere during thermal inversions. The ambient concentrations of total suspended particulates (TSPs) in major urban locations have regularly exceeded World Health Organization (WHO) guidelines.¹ The number of days with a visibility of more than 8,000 m (at 11:45 hr local time) during the period from November to February, the months with lowest visibility, has decreased from 115 days in 1970 to only about 20 days in 1993.² With rapidly growing fuel use due to increasing urbanization and industrialization, air quality is likely to deteriorate further if no effort is made to improve the situation.

There exist a few studies on air-quality measurements,¹⁻⁴ which are mainly focussed on ambient air-quality issues while others focus on estimation of air pollutants related to fuel use from a few sectors.⁵⁻⁷ It is of interest to identify the major pollutants for all sectors so that effective mitigation programs and policies may be formulated.

2. METHODOLOGY AND DATA SOURCES

The anthropogenic emissions arise from mobile (road vehicles) and stationary (households, industries and commercial enterprises) sources. Here, we estimate sectoral emissions of CO, hydrocarbons (HCs), nitrogen oxides (NO_x), sulfur dioxide (SO₂), and TSPs.

2.1. Mobile sources

We define $F_i(t)$ = average fuel consumption by vehicle type i in year t , $FE_i(t)$ = fuel efficiency of vehicle type i in year t , $vkmi(t)$ = average vehicle-km travelled by vehicle type i in year t , $N_i(t)$ = number of vehicles in operation by vehicle type i in year t , and $M_{ij}(t)$ = exhaust emission by vehicle of type i for pollutant type j in year t .

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‡Author to whom all correspondence should be addressed.

Since fuel efficiency is a function of speed and vehicle age,⁸ the average vehicle-km travelled by vehicle type i in year t is

$$vkm_i(t) = F_i(t) FE_i(t) SCF_i(t) ACF_i(t), \quad (1)$$

where $SCF_i(t)$ denotes a speed-correction factor (defined as the ratio of the pollutant-exhaust-emission rate at any speed to the pollutant-exhaust-emission rate at a specified speed, as determined by the 1975 Federal Test Procedure) for vehicle of type i in year t and $ACF_i(t)$ represents an age-correction factor (defined as the ratio of pollutant-exhaust-emission rate at any vehicle-use status in km to the pollutant-exhaust-emission rate at a specified km). $ACF_i(t)$ is used to adjust for deterioration of vehicle performance with vehicle age. The exhaust emission by vehicle type i for pollutant type j in year t is

$$M_{ij}(t) = N_i(t) vkm_i(t) EF_{ik}, \quad (2)$$

where EF_{ik} is the exhaust-emission factor (defined as the mass of pollutant emitted per unit of distance travelled). Estimated values of average fuel consumption, fuel efficiency and number of vehicles in operation are presented in Table A.1 in the Appendix while Table A.2 shows emission factors used for mobile sources.

2.2. Stationary sources

Let $F_{ks}(t)$ = fuel consumption of type k used in sector s in year t , EF_{jks} = emission factor of pollutant type j for fuel type k used in sector s , and $P_{jks}(t)$ = emission of pollutant type j for fuel type k used in sector s in year t . The emission of pollutant type k from the use of fuel type j in sector s in year t is estimated to be

$$P_{jks}(t) = F_{ks}(t) EF_{jks}. \quad (3)$$

The shares of fuel consumption for stationary sources are shown in Table 1. Emission factors for stationary sources equal the masses of pollutants emitted per unit of fuel consumed. Table A.3 in the Appendix presents emission factors for stationary sources.

Table 1. Estimated sectoral shares (%) of fuel consumption in Kathmandu Valley in 1993.⁹

Fuel Type	Household		Industrial			Transport	Commercial	Total
	Urban	Rural	Brick	Cement	Others [†]			
Biomass fuels								
Animal wastes	-	0.2	-	-	-	-	-	0.2
Agricultural residues	0.7	6.6	0.5	-	0.7	-	-	8.5
Fuelwood	11.6	10.8	3.8	-	3.2	-	<0.0	29.2
Charcoal	0.2	-	0.7	-	0.1	-	0.2	1.2
Fossil fuels								
Coal	-	-	14.6 ⁵	4.9	1.4	-	-	20.8
Kerosene	9.3	5.5	<0.0	-	<0.0	-	0.3	15.2
Light diesel oil	-	-	0.3	-	-	10.1	-	0.3
High speed diesel	-	-	-	-	-	11.4	-	10.1
Gasoline	-	-	-	-	-	-	-	11.4
LPG	2.3	0.3	-	-	-	-	0.2	2.8
Total	24.0	23.3	19.9	4.9	5.5	21.5	0.9	100.0

[†] Other industrial activities include industries such as food, textile, wood, paper, chemicals, machinery, leather, and miscellaneous based on the Nepal Industrial Standard Code (NISC).

2.3. Data sources

Vehicular data (i.e. fuel consumption, fuel efficiency, speed, and registration date) were collected from a sample survey of 194 vehicles of different types. Socio-economic and energy-use data related to urban households were gathered from a sample survey of 216 randomly selected households. Data on rural households and on industrial and commercial sectors are based on Refs. 10 and 11, respectively.

3. FUEL-USE PATTERNS

Nepal has no indigenous fossil fuels. The use of fossil fuels imposes heavy pressure on the trade balance, with oil imports alone requiring about one third of total export earnings.¹² Almost half of the country's total fossil-fuel imports in 1993 were consumed in Kathmandu Valley. In 1993, total fuel consumption was about 203 ktce. Sectoral fuel shares in 1993 are presented in Table 1.

As may be seen from Table 1, the household sector had the largest share of total fuel consumption, followed in descending order by the industrial, transport and commercial sectors. The household sector accounts for almost four fifths of the total biomass fuels used. The industrial and transport sectors account for more than two thirds of total fossil-fuel consumption. Fuelwood, coal and kerosene are the dominant fuels. Sectoral fuel use will now be summarized.

3.1. Household sector

Biomass and kerosene are the main fuels. However, there are indications of rapidly growing substitution of biomass by fossil fuels.^{10,13} The weighted average per capita of biomass-fuel consumption in 1993 was lower than in 1985, while the opposite was observed for modern fuels during this period. Table 2 shows estimated values of average annual per capita fuel consumption by urban and rural households. Urban households are found to use less traditional primary energy per capita and more modern energy per capita than rural households, primarily due to heavy dependence on biomass fuels and inefficient appliances in rural areas.

3.2. Industrial sector

Coal (78%) and fuelwood (17%) are the dominant fuels. Brick making accounts for about two thirds of total fuel consumption and is the largest emission source; it is a relatively inefficient technology with an average specific fuel requirement for fired brick (one fired brick weighs approximately 2 kg) production ranging from 400–500 kcal/kg as compared to 245 kcal/kg for bricks produced with vertical shaft brick kilns (VSBK) technology.¹⁴ The shares of the cement and other industries of total industrial fuel consumption are 16 and 18%, respectively.

3.3. Transport sector

All high-speed diesel and gasoline uses refer to the transport sector. The registered vehicle population has almost doubled during the last decade.¹⁵ The average fuel efficiency of cars (i.e. 10.6 km/l) in 1993 was much lower than for Japanese, Taiwanese and German cars of comparable size in 1985.¹⁶ The low average fuel efficiency partly reflects bad road conditions, poor maintenance and large fractions

Table 2. Estimated per capita consumption of fuels in urban and rural areas in Kathmandu Valley in 1993 (kgce). Figures in parentheses represent useful energy consumption per capita (kgce).[†]

Area	Agricultural residues	Animal wastes	Fuelwood	Kerosene	Charcoal	LPG	Total
Urban	2.18 (0.26)	- -	36.46 (4.38)	29.05 (14.53)	0.55 (0.12)	7.18 (4.67)	75.45 (23.96) [‡]
Rural	25.11 (3.01)	0.67 (0.32)	40.76 (4.89)	20.84 (10.42)	- -	1.03 (0.67)	88.44 (19.31)

[†] The assumed values of appliance efficiencies are 0.12 for traditional stoves (i.e. stoves using agricultural residues, animal wastes, and fuelwood), 0.22 for charcoal stoves, 0.50 for kerosene stoves, and 0.65 for LPG stoves.¹² [‡] The corresponding figure in 1990 was 33.5 kgce for Beijing¹⁷ and 94–168 kgce for Hongkong.¹⁸

of old and/or reconditioned cars. Vehicle ownership (47 vehicles per 1000 persons in 1990) is among the lowest for cities in LDCs. Corresponding figures for Bangkok, Bombay, Jakarta, Karachi, and Mexico City were 246, 58, 165, 95, and 137 per 1000 persons, respectively.¹⁹

4. EMISSION OF AIR POLLUTANTS

4.1. Air quality in the valley

Limited studies that exist on air quality indicate that there is severe street-level air pollution. In many parts of the valley, TSP concentrations exceed WHO guidelines and are comparable with those in Mexico City (with an average of 100 to 500 $\mu\text{g}/\text{m}^3$ in 1992),¹⁹ while CO, SO₂ and NO₂ concentrations are lower. Table 3 shows the ambient concentrations of selected pollutants.

4.2. Emission inventory for air pollutants

Table 4 shows an inventory of sectoral emissions of TSPs, CO, HCs, NO_x, and SO₂. The transport sector made the largest contribution, followed by the household, industrial and commercial sectors in descending order.

Total emissions from fossil fuels are much greater (almost 3 times) than from biomass fuels. Gasoline is the largest contributor (55%) to the pollutants, while fuelwood and coal are other major contributors with shares of 16 and 14%, respectively. Gasoline contributed 59 and 70% of total CO and HC emissions, respectively. Gasoline, fuelwood, coal, and agricultural residues emitted about 93% of the total. Table 5 shows emissions of pollutants by fuel type in 1993. The sectoral emission of pollutants will now be summarized.

Transport sector. Total vehicular exhaust emissions were over 36,000 tons, with CO contributing 65%, HCs 30%, and NO_x 3%. The amounts of TSPs and SO₂ were very small. Total emissions of all pollutant types per vehicle were 0.76 ton in 1993;† 2-wheelers contributed 56%, cars 36% and 3-wheelers 3%, while other vehicles accounted for about 5%. The high share of 2-wheelers is expected, as these represent over two thirds of total vehicular registration and also have low average fuel efficiency (45.5 km/l).‡ The contribution from 3-wheelers (in particular, diesel-operated, India-made Vikram

Table 3. Ambient air-quality concentration ($\mu\text{g}/\text{m}^3$) in Kathmandu Valley in 1993.¹⁹

Site	24-hour Concentration				
	TSPs	PM ₁₀ [†]	SO ₂	NO ₂	CO [‡]
Commercial (traffic) sites					
Average	187 - 391	67 - 142	17 - 77	11 - 37	-
Maximum	319 - 867	86 - 201	64 - 202	16 - 64	-
Residential sites					
Average	191 - 228	72 - 113	13 - 110	12 - 49	-
Maximum	270 - 350	126 - 161	13 - 225	14 - 126	-
Industrial sites					
Average	87 - 430	40 - 166	13 - 58	20 - 40	-
Maximum	102 - 560	53 - 194	13 - 79	24 - 83	-
WHO guidelines					
Average	150 - 230	100 [§]	100 - 150	150	30,000 [¶]

[†] PM₁₀ is the fraction of TSPs less than 10 microns in diameter. Essentially all particulates derived from combustion are less than 10 microns in size, while particulates derived from mechanical action such as road dust are larger.

[‡] The average value ranges from 18,750 to 25,000 $\mu\text{g}/\text{m}^3$ for a 1-hour average.⁵ [§] This is the 24-hour time-weighted average for Japan. [¶] This is a 1-hour average.

†This value seems to be very high as compared to that of the major Indian cities, i.e. Bangalore, Bombay, Calcutta, Delhi, and Madras with the corresponding figures of 0.20, 0.34, 0.22, 0.19, and 0.22 ton, respectively, in 1982.²⁰

‡The average fuel efficiencies for 2-wheelers were 62.5 km/l for 4-stroke engine models in Thailand and 51.0 km/l for the 1992 model with less than 125 cc in the USA.²¹

Table 4. Estimated total emissions of selected pollutants from energy use in Kathmandu Valley in 1993 (tons).† Figures in parentheses are shares (in %) of pollutants in the total emissions.

Source	TSPs	CO	HCs	NO _x	SO ₂	Total
Trucks	86	346	107	375	50	964
Buses	8	33	10	36	5	92
Minibuses	19	28	16	160	5	228
Jeeps	59	207	86	92	25	469
Tractors	2	39	22	24	7	94
Cars	38	11,178	1,496	487	23	13,222
3-wheelers	29	609	379	146	5	1,168
2-wheelers	234	11,253	8,908	33	9	20,437
Sub total (Transport [‡])	475	23,693	11,024	1,353	133	36,678
Urban	1,028	3,672	500	112	139	5,451
Rural	1,354	6,195	781	101	364	8,795
Sub total (Household)	2,382	9,867	1,281	213	503	14,246
Brick						
Bull Trench	2,018	3,088	558	359	774	6,797
Chinese	328	410	91	67	164	1,060
Cement	615	769	171	127	308	1,990
Others	613	953	672	75	103	2,416
Sub total (Industrial)	3,574	5,220	1,492	628	1,349	12,263
Commercial	24	234	11	5	3	277
Total	6,455 (10.2)	39,014 (61.5)	13,808 (21.7)	2,199 (3.5)	1,988 (3.1)	63,464 (100.0)

† These values are likely to underestimate the total emissions from fuel use since they do not include emissions from unregistered commercial enterprises, cottage industries and open refuse burning. Also, emissions of lead from households, industries and the commercial sector are not included because of lack of data. Emissions of lead have been estimated only for the transport sector and are 4 tons, 1 ton, and 1 ton from cars, 3-wheelers (gasoline), and 2-wheelers, respectively. ‡ Figures refer only to exhaust tailpipes and do not include crankcase blowby and evaporative emissions. About 20% of the TSPs emitted by vehicles are from crankcase blowby while 20% of the total HC emissions are evaporative.²⁰

tempos) was relatively small since these constitute only about 5% of the total number of registered vehicles and have lower emissions of CO and HCs than gasoline-powered vehicles. This finding should not be viewed as contradicting the popular view that 3-wheelers are mainly responsible for “blue smoke” in the city center, since they may be major polluters at the street level during peak hours.

Our estimates suggest that the combined emission of CO and HCs from vehicles accounted for almost 95% of total vehicular emissions. Although gasoline consumption was only about 28% greater than diesel consumption, the former contributed 94% of the total vehicular emissions. This result is mainly due to a substantially higher rate of emissions of CO and HCs for gasoline vehicles than for diesel vehicles.† For example, the specific emissions of CO and HCs from gasoline vehicles are 778 and 361 kg/kl, respectively, compared to 29 and 11 kg/kl from diesel vehicles.²³

Household sector. The total emissions from this sector in 1993 were over 14,000 tons, of which more than two thirds was CO. The shares of TSPs and HCs were 17 and 9%, respectively. The total rural household emissions (62%) are higher than urban household emissions (38%). Biomass fuels contributed about 89% of total emissions from this sector. In per capita terms, emissions from rural households of pollutants were significantly higher than for urban households; per capita emissions of CO by rural and urban households were 11.6 and 5.7 kg, respectively, while the corresponding TSP emissions were 2.5 and 1.6, respectively.

Industrial sector. Brick making accounted for two thirds of sectoral total emissions. Most (55%)

†Exhaust emissions from gasoline vehicles consist mainly of CO, HCs and NO_x, while emissions from diesel vehicles have low CO and HCs but high NO_x concentrations.²²

Table 5. Total emissions by fuel type from mobile and stationary sources in Kathmandu Valley in 1993 (tons).

Fuel Type	Sector	TSPs	CO	HCs	NO _x	SO ₂	Total
Animal wastes	Household	14	72	11	1	9	107
Agricultural residues	Household	513	3,843	381	35	307	5,082
	Industrial	238	287	41	10	28	604
Fuelwood	Household	1,748	4,660	874	81	70	7,432
	Industrial	661	1,047	697	70	3	2,478
	Commercial	7	18	3	<1	<1	29
Charcoal	Household	10	129	5	1	<1	146
	Industrial	48	603	24	5	1	681
	Commercial	15	183	7	2	<1	207
Sub total (Biomass fuels)		3,254	10,842	2,046	204	420	16,766
Coal	Industrial	2,627	3,283	730	540	1,313	8,493
Kerosene	Household	97	1,052	5	69	118	1,341
	Industrial	<1	<1	<1	2	1	3
	Commercial	2	23	<1	2	3	30
Light diesel oil	Industrial	<1	<1	<1	<1	3	5
High speed diesel	Transport	192	677	254	829	96	2,046
Gasoline	Transport	285	23,016	10,770	525	34	34,630
LPG	Household	<1	110	1	26	<1	137
	Commercial	<1	9	<1	2	<1	12
Sub total (Fossil fuels)		3,201	28,172	11,762	1,995	1,568	46,698
Total		6,455	39,014	13,808	2,199	1,988	63,464

emissions resulted from Bull Trench (BT) type of brick making. Coal and fuelwood contributed over three quarters and one sixth of total emissions, respectively.

Commercial sector. The contributions from this sector to total emissions were negligible.

4.3. Pollutant mix

CO was predominant (61%) among the pollutants emitted, followed by HCs (22%), TSPs, NO_x, and SO₂. The transport and household sectors contributed most to CO and HCs emissions, while the industrial sector contributed most to CO and TSPs emissions. It should, however, be noted that although the emission of TSPs from fuel use appears to be relatively small, it is likely that large quantities of TSPs are also emitted from non-energy sources such as poor road quality, construction activities, rice and saw-milling, brick-farming production and lack of regular road cleaning. Estimations of TSP emissions from non-energy sources are beyond the scope of the present paper.

Per capita emission and emission density (defined as emissions per unit area) of TSPs and CO in 1990 are higher than for many other cities in developing countries. The per capita emissions of TSPs in the valley were 5.47 kg while they were 4.49 kg in Bombay, 0.34 kg in Buenos Aires and 4.18 kg in São Paulo. Similarly, per capita CO emission was 36.06 kg compared with 16.89 kg in Bombay, 20.73 kg in Buenos Aires and 14.96 kg in Calcutta. The emission densities of TSPs, CO, and NO_x were, respectively, 21.5, 130.0 and 7.3 tons/km², whereas they were 18.0, 118.0 and 7.0 tons/km², respectively, in Mexico City.¹⁹

4.4. Future emissions under the BAU scenario

Emissions in 2013 were estimated for the following sectoral assumptions. *Transport sector*—gasoline and diesel use grow, respectively, at average annual growth rates (AAGRs) of 8.5 and 8.9% from 1993 to 2013.¹² The percentage shares of 2-wheelers, cars and 3-wheelers of total gasoline consumption in

2013 will be 45, 40 and 15, respectively, while the shares of diesel vehicles for total diesel consumption are the same as in 1993. *Household sector*—the AAGRs for urban and rural populations will be 4.50 and 2.75%, respectively.²⁴ The per capita fuel consumption in 2013 will be the same as in 1993. *Industrial sector*—production of bricks in 2013 equals the product of population in 2013 and brick requirement per capita as in 1993. Local production of cement will be at the same level as in 1993. Fuel requirements for other industrial activities by 2013 will grow at the same rate as the industrial GDP from 1981–1991.²⁵ *Commercial sector*—fuel requirements for the sector by 2013 will grow at the same rate as the commercial GDP from 1981–1991.²⁵

Estimated values of sectoral emissions in 2013 are shown in Table 6. The combined emissions of all pollutants in 2013 under the specified scenario are estimated to be about five times those in 1993. More than two thirds of the increase will be contributed by the transport sector, about one sixth by the industrial sector and about 10% by the household and commercial sectors. In the absence of mitigation measures, the per capita emission of all pollutants combined in the year 2013 will be more than three times that of 1993.

Table 6. Sectoral emission (in tons) for Kathmandu Valley in 2013 under the BAU.

Sector	TSPs	CO	HCs	NO _x	SO ₂	Pb	Total
Transport	3,441	141,981	90,238	6,015	693	32	242,400
Household	4,809	19,512	2,549	443	963	-	28,275
Industrial	11,113	16,542	4,488	1,943	4,143	-	38,229
Commercial	54	535	25	12	7	-	634
Total	19,417	178,570	97,300	8,413	5,806	32	309,538

5. CONCLUSIONS

The total emission of all selected pollutants in 1993 was more than 63,000 tons, of which about two thirds was CO; the transport sector was the largest sectoral contributor while gasoline was the dominant fuel contributor to the total. In the transport sector, 2-wheelers and cars accounted for over 92% of the total emissions. Rural households (despite their lower share of the total population) were found to have a larger share in household-sector emissions than urban households because of the heavy dependence on biomass. Among industries, brick making contributed most to pollution.

Under the BAU scenario, total emissions in 2013 are estimated to be about five times those of 1993. About two thirds of the increase will come from the transport sector. Because of the assumptions made, these estimates should not be treated as precise values.

There is a large potential for mitigation of major air pollutants through improvements in fuel quality and energy efficiency, use of emission-control technologies, and efficient fuel pricing. Introductions of electric vehicles and of mass-transport systems appear to be useful options for mitigating emissions from the transport sector. In the household sector, inter-fuel substitution and introduction of improved cooking stoves (ICS) should be considered. Efficient VSBK technology (similar to that widely used in China) appears to be a promising option in the industrial sector.

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APPENDIX

*Vehicular parameters and emission factors used in the study*Table A.1. Estimated average fuel consumption, fuel efficiency, number of operating vehicles, and km travelled by vehicle type in Kathmandu Valley.⁹

Fuel Type	Vehicle Type	Average Fuel Consumption (l)	Fuel Efficiency (km/l)	Number of Operating Vehicles	Average km Travelled Per Vehicle [†]
Diesel	Truck	8,704	4.5	736	39,168
	Bus	8,418	3.0	110	25,254
	Minibus	7,373	4.5	372	33,178
	Jeep	2,315	8.0	3,555	18,520
	Tractor	4,785	4.4	822	21,054
	3-Wheeler	2,592	12.5	335	32,400
Gasoline	Car	1,595	10.6	10,664	16,907
	3-Wheeler	1,479	11.0	1,588	16,269
	2-Wheeler	341	45.5	30,220	15,515

[†] Figures are derived assuming that both the speed and age correction factors are 1 for all vehicle types.⁸

Table A.2. Emission factors for mobile sources (g/km).^{5,23,26,27}

Fuel Type	Vehicle Type	TSPs	CO [†]	HCS [†]	NO _x	SO ₂ [†]	Pb [‡]
High speed diesel	Truck	3.00	12.00	3.70	13.00	1.75	-
	Bus	3.00	12.00	3.70	13.00	1.75	-
	Minibus [§]	1.50	2.25	1.26	13.00	0.39	-
	Jeep	0.90	3.10	1.30	1.40	0.38	-
	Tractor	0.90	2.25	1.26	1.40	0.39	-
	3-wheeler	1.50	2.25	1.26	13.00	0.39	-
Gasoline	Car	0.20	62.00	8.30	2.70	0.13	0.02
	3-wheeler	0.21	22.64	14.13	0.20	0.05	0.02
	2-wheeler	0.50	24.00	19.00	0.07	0.02	0.003

[†] Figures for trucks, buses, and cars are based on urban driving conditions (24 km/hr) for uncontrolled vehicles. Figures for jeeps and 2-wheelers are based on driving conditions in Medan City (Indonesia) during 1993 for uncontrolled vehicles.²⁶ [§] Figures for the minibus are assumed to be the same as for a tractor and for diesel-operated 3-wheeler.²³ [‡] Gasoline contains 0.18 g/l of the lead; 70% of the lead supplied to an engine is released from the exhaust tailpipe.²⁷

Table A.3. Emission factors for stationary sources (g/kg).^{5,9,12,27-29}

Fuel Type	Sector	TSPs	CO	HCS	NO _x	SO ₂
Animal Wastes	Household	10.0	50.0	7.5	0.7	6.0
Agricultural Residues	Household	10.0	75.0	7.5	0.7	6.0
	Industrial	29.0	35.0	5.0	1.2	3.4
Fuelwood	Household/Commercial	15.0	40.0	7.0	0.7	0.6
	Industrial	18.0	28.5	35.7	1.9	0.07
Charcoal	Household/Commercial	20.0	250.0	10.0	2.25	0.39
	Industrial [‡]	20.0	250.0	10.0	2.25	0.39
Coal	Industrial	36.0	45.0	10.0	7.4	18.0
Kerosene [†]	Household/Commercial	3.5	38.0	0.2	2.49	4.25
	Industrial	0.69	0.63	0.12	7.46	5.0
Light diesel oil	Industrial	0.3	0.63	0.12	2.4	6.0
LPG [†]	Household/Commercial	0.1	24.0	0.2	5.25	0.02

[†] The average maximum sulfur contents of kerosene and LPG are 0.25 and 0.02%, respectively, by weight on a dry basis. [‡] The figures are assumed to be the same as for the household sector.